# Thomas Fieguth

Robotics & Mechatronics Engineer
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## Education

**BASc Systems Design Engineering (Co-op)** – University of Waterloo · *Graduated with Dean's Honours* Apr 2026

## Skills

- **Mechanical:** SolidWorks, Onshape, CNC (mill & lathe), machining, welding, soldering, woodworking
- **Programming:** C++, Python, Kotlin, Onshape FeatureScript, Linux, Git
- **Robotics:** ROS2, sensor fusion (EKF), path planning & tracking, point clouds / ICP, Qt
- **Languages:** English and German (fluent), French (immersion)

## Experience

**Werkr Tractors** – Waterloo, ON Jan 2024 – Present
*Software Engineer (Full-time), May 2026 – Present · Software / Robotics Co-op, Jan – Apr 2024 and May – Aug 2025*

- Designed the autonomous tractor's ROS2 guidance system: a mission-manager state machine, a speed-profiled trajectory planner, and a modified Stanley controller with curvature feedforward
- Built path planning for 5 mission types with smoothed Dubins/Reeds-Shepp paths to avoid abrupt steering changes during turns
- Tuned the Extended Kalman Filter fusing GPS position, GPS heading, and wheel odometry; wrote Python ROS2 bag tooling that found and fixed 3 estimator faults (GPS overconfidence, heading overweighting, yaw jumps)
- Added Nav2 obstacle handling with custom costmaps, in-row detours, and a collision monitor
- Built a Qt/ROS2 operator interface for live tractor position, state feedback, and path generation

**GEI Consultants, Inc.** – Kitchener, ON May – Aug 2023
*Project Designer Co-op*

- Analyzed pump efficiency and friction losses at wastewater pumping stations, showing the system ran at 73% of expected capacity
- Built a parametric cost model for new water/wastewater treatment plants using only flow rate and process type
- Conducted site inspections and wrote formal client reports

**Accelerated Systems Inc.** – Waterloo, ON Jan – Dec 2022
*Mechanical / Software Engineer Co-op, Sep – Dec 2022 · Machinist & Technical Support Co-op, Jan – Apr 2022*

- Designed, built, and tested a water-cooling jacket for an electric motor
- Designed in SolidWorks and built a motor test stand and dynamometer for e-bike motors
- Programmed CNC machines for prototyping; machined and assembled airflow-optimized golf cart motor casings
- Built a Kotlin app to remotely drive an autonomous lawnmower with live object detection and sensor readouts, and wrote C++ ROS2 nodes for its proximity safety stops, geofencing, and joystick teleoperation

## Projects

**Depth-Sensing Woodcarving Guide** – *Capstone, team of 4* Sep 2025 – Apr 2026
*Winner, N.S. Robertson Foundation Capstone Design Award*

- Built a depth camera and projector rig that compares point clouds to the user's CAD model and projects a live depth-difference map onto the wood, showing where and how deep to carve next
- Owned all CAD and mechanical fabrication: 3 frame prototypes and 5 camera-projector mount designs
- Helped build and debug the camera-to-projector pipeline; developed 3 iterations of the Python/Qt interface

**Personal Projects**

- **Farm Topography Relief Map:** Onshape FeatureScript tool that turns elevation data into nested, self-supporting CNC contour layers in 9 wood species
- **Catan River Tile Optimizer:** Monte Carlo search with DFS conflict resolution to maximize valid river layouts
- **Stormbreaker Axe Replica:** load-bearing 1/4" steel skeleton under a sheet-metal shell; welded and heat-treated
- **CNC Wooden Globe:** 60 CNC-cut walnut and maple triangles assembled into a hollow, motor-driven globe